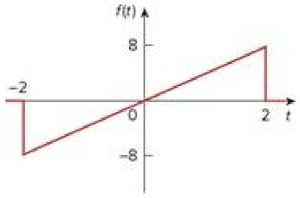
Chapter 18, Problem 3P

Problem

Calculate the Fourier transform of the signal in Fig. 18.28.

Figure 18.28



Step-by-step solution

Step 1 of 3

Refer to the signal $f(t)$ in Figure 18.28 in the textbook.

The signal $f(t)$ is expressed as,

$$f(t)=\begin{cases}4t, & -2\le t\le 2\\ 0, & \text{otherwise}\end{cases}$$

Step 2 of 3

The Fourier transform of the signal $f(t)$ is,

$$\begin{aligned}F(\omega)&=\int_{-\infty}^{\infty} f(t)e^{-j\omega t}dt\\&=\int_{-\infty}^{-2} f(t)e^{-j\omega t}dt+\int_{-2}^{2} f(t)e^{-j\omega t}dt+\int_{2}^{\infty} f(t)e^{-j\omega t}dt\\&=\int_{-\infty}^{-2}[0]e^{-j\omega t}dt+\int_{-2}^{2}[4t]e^{-j\omega t}dt+\int_{2}^{\infty}[0]e^{-j\omega t}dt\\&=\int_{-2}^{2}4te^{-j\omega t}dt\end{aligned}$$

Step 3 of 3

Simplify further.

$$
\begin{aligned}
F(\omega) &= 4\int_{-2}^{2} te^{-j\omega t}\,dt \\
&= 4\left[\frac{e^{-j\omega t}}{(-j\omega)^2}(-j\omega t-1)\right]_{-2}^{2} \\
&= \frac{-4}{\omega^2}\left[e^{-j\omega t}(-j\omega t-1)\right]_{-2}^{2} \\
&= \frac{-4}{\omega^2}\left[e^{-j2\omega}(-j2\omega-1)-e^{j2\omega}(j2\omega-1)\right]
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{-4}{\omega^2}\left[-j2\omega\left(e^{-j2\omega}+e^{j2\omega}\right)+e^{j2\omega}-e^{-j2\omega}\right] \\
&= \frac{-4}{\omega^2}\left[-j2\omega(2\cos 2\omega)+2\sin 2\omega\right] \\
&= \frac{8}{\omega^2}\left[j2\omega\cos 2\omega-\sin 2\omega\right]
\end{aligned}
$$

Therefore, the Fourier transform of signal $f(t)$ is $\boxed{\frac{8}{\omega^2}\left[j2\omega\cos 2\omega-\sin 2\omega\right]}$.